

PROJECT TITLE SHOULD BE IN UPPERCASE
LETTERS & TIMES NEW ROMAN

A PROJECT REPORT
BY

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SUBMITTED TO

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND
TECHNOLOGY, BENNETT UNIVERSITY
GREATER NOIDA, 201310, UTTAR PRADESH, INDIA

April 2026

DECLARATION

I/We hereby declare that the work which is being presented in the report entitled “Project Title”, is an authentic record of my/our own work carried out during the period from JAN, 2023 to April, 2023 at School of Computer Science and Engineering and Technology, Bennett University Greater Noida.

The matters and the results presented in this report has not been submitted by me/us for the award of any other degree elsewhere.

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ACKNOWLEDGEMENT

I/We would like to take this opportunity to express my/our deepest gratitude to my/our mentor, **Dr. IJKL (Provide correct name & designation)** for guiding, supporting, and helping me/us in every possible way. I/we was/were extremely fortunate to have him as my/our mentor as he provided insightful solutions to problems faced by me/us thus contributing immensely towards the completion of this capstone project. I/We would also like to express my/our deepest gratitude to VC, DEAN, HOD, faculty members and friends who helped me/us in successful completion of this capstone project. Any other name you can mentioned here. (Acknowledgement is your personal view, so you can write it in your way by maintaining integrity of technical report).

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LIST OF ABBREVIATIONS

Abbreviation.....	Explanation of the Abbreviation
AAA Control	Authentication Authorization and Access
CSP	Cloud Service Provider
DNS	Domain Name System
IAM	Identity and Access Management

ABSTRACT

Abstracts must use 250 words to 500 Basic guidelines for the preparation of a technical work for the project report. This document is itself an example of the desired layout (inclusive of this abstract) and can be used as a template. The document contains information regarding desktop publishing format, type sizes, and typefaces. Style rules are provided that explain how to handle equations, units, figures, tables, abbreviations, and acronyms. Sections are also devoted to the preparation of acknowledgments, references, and authors' biographies. The abstract is limited to 500 words and cannot contain equations, figures, tables, or references. It should concisely state what was done, how it was done, principal results, and their significance.

1. INTRODUCTION

<<Provide a brief overview of the current trends and situation around your project>>

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- Bulleted list items use the “Bullets” style from the styles pane.

1.1. Problem Statement

<< Provide a concise statement on the problem that currently exists and is affecting the organization/society/task.

Example: Currently, Microsoft uses Microsoft Dynamics AX as their enterprise resource planning (ERP) solutions for businesses. The latest version of Microsoft Dynamics AX will collect a significant amount of telemetry about the user’s actions while navigating in the application and its thousands of forms.>>

2. BACKGROUND RESEARCH

<<Provide a detailed description of the literature research (**atleast 2 pages**) that you conducted around your project topic (make sure to cite the relevant sources in your text. E.g., research papers/web articles/blogs) and what motivated you, after researching, to work on the project.

Would suggest you to download a tool (Mendeley: <https://www.mendeley.com>) for automated citation and generation of references.>>

2.1. Proposed System

<< Provide a concise statement on the purpose of the project; the problem or opportunity addressed. The explanation should include what you intended to do. Vision – how will the customer’s world improve as a result of this project? When appropriate, tie this into what is currently being considered or has just been completed at the organization

Example: This project aims to extract information about how users use the forms by analyzing the raw telemetry data. By using big data analysis and machine learning techniques we hope to develop predictions into what actions the user will do next. Using these insights, the Microsoft Dynamics AX engineering may be able to optimize the user experience and reduce the number of steps needed to perform desired actions or reach desired forms.>>

2.2. Goals and Objectives

<< State, in quantifiable terms, if possible, the goals and objectives of the project. Goals may be related to product, process, quality, or teamwork. >>

Example:

<<to insert caption for tables and figures, click on references on the top menu, under captions section → click insert caption → choose label as table>>

Table 1: Goal and Objectives

#	Goal or Objective
1	Make the system extensible – future updates like xxx can be done easily
2	Make the system easy to support – provide good documentation, configuration/build files, administrator’s manual
3	Make the system very easy to use – users would agree that minimal to no training is needed
4	Build a prototype that demonstrates the user interface by xx/xx/xx - in order to get early feedback from the customer/users
5	Have fun working on the project

3. PROJECT PLANNING

This section covers the details of the project planning. Selecting the lifecycle of the development, project stakeholders, resources required, assumptions made (if any) are detailed in the sections below.

3.1. Project Lifecycle

<< Describe the lifecycle of the project. You can choose from an existing lifecycle definition or create your own.

Example: The team will use an agile approach. Our team will gather requirements and create a high level development plan at the onset of the project and then implement the gathered requirements over three iterations. The team will follow a SCRUM-like approach with an emphasis on frequent meetings and collaboration.>>

3.2. Project Setup

<< Define some of the basic project decisions that will be used on this project.

>>

Example:

Table 2: Sample 2

#	Decision Description
1	Windows 8, C#, OpenSphere vs. Azure, Trac/SVN vs. Git, etc.
2	Standards that must be followed (default Capstone coding standard, etc.)
3	Special access privileges needed, nondisclosure forms, release to open source, etc.
4	A virtual server image will be set up at BU that matches the customer environment (image provided by customer)

3.3. Stakeholders

<< Identify all stakeholders for this project (groups or individuals that are affected by or are in some way accountable for the outcome of the project – business managers, end users, developers, testers, support people, instructors, etc.)

>>

Example:

Table 3: Sample 3

Stakeholder	Role
Person A	Sponsor
Person B	Mentor

Person C	Instructor
Person D	Team member
Person E	Team member
Person F	Team member

3.4. Project Resources

<< Identify the anticipated resources required for this project. This can include staff members who will work on the project, equipment needed for the project, special software that will need to be acquired, or any other resource necessary for the project. >>

Example:

Table 4: Sample 4

Resource	Resource Description	Quantity
Database Server	A database server provided by the sponsoring company.	1
Capstone Team	Our team of students who will be the primary developers of the project.	4
Jim Somebody	The mentor who will be able to provide us with technical assistance.	1
Mac Workstation	An OS X workstation with X Code for developing the OS X version of the software.	1
Android Phone	An Android phone to be used as test hardware for the mobile version of the software.	2

3.5. Assumptions

<<State any assumptions upon which the project is based. Assumptions may be related to staffing, resources, tools, and schedules/deadlines. >>

Example:

Table 5: Sample 4

#	Assumption
A1	The capstone team and mentors will be able to meet face to face once a week.
A2	Azure ML will be available for the team to work with as a trial for the first month of the project.
A3	Team members will be able to familiarize themselves with the Azure ML, Azure HDInsights, and R environments
A4	Team will have sufficient time to complete a working model to present by mid-semester
A5	Machine Learning model will be completed in time to test on true big data using HDInsights and Hadoop
A6	The development test data provided will be sufficient to create an accurate prediction of user actions
A7	The models developed will be easily extended to other forms within the time frame

4. PROJECT TRACKING

4.1. Tracking

<< Provide information about how the project was tracked and where information was kept. This should include information such as what type of source control was being used and how it can be accessed, any bug-tracking system that was used for the project and where it can be accessed, what type of regression testing suite was used and where it can be accessed, and any similar information that provides details on the project's status, etc. >>

Example:

Table 6: Sample 6

Information	Description	Link
Code Storage	Project code will be stored in SVN repository.	Link
Bug Tracking	Bug tracking will be done with Trac.	Link
Project Documents and Assignments	Weekly reports, specification and design documents, etc. will be stored in our SVN repository.	Link
Continuous Integration	Continuous integration will be done with Jenkins.	Link
Regression Testing	Regression testing will use JUnit unit tests and Jenkins.	Link

4.2. Communication Plan

<< Identify all communications you will provide to other groups and all communications you need to receive from other groups. Share this information with affected groups. Verify that all stakeholders are included. >>

Table 7: Regularly Scheduled Meetings

Meeting Type	Frequency/Schedule	Who Attends
Conference Call/Skype	Weekly	Project team and mentor
Team Meeting	Weekly	Project team
Short Meeting	Weekly in class	Project team
Sprint Planning Meeting	Start of each sprint	Project team and mentor
Sprint Retrospective Meeting	End of each sprint	Project team
Sprint Review Meeting	End of each sprint	Project team, <i>mentor, and sponsor</i>

Table 8: Information To Be Shared Within Our Group

Who?	What Information?	When?	How?
Project team	Task assignments & General scrum information	Weekly	Team meetings, listing in Project Specification.

Table 9: Information To Be Provided To Other Groups

Who?	What Information?	When?	How?
Sponsor and mentor	Final deliverables	At completion of project	Project specification doc., code, Power Point presentation
Sponsor and mentor	Weekly report	Weekly	Email and Trac site access
Sponsor and mentor	Project baselines <i>(optional)</i>	At the end of each sprint	Onsite customer demo, access to repository

Table 10: Information Needed From Other Groups

Who?	What Information?	When?	How?
Sponsor and mentor	Requirement changes	Start of each sprint	Conference call or meeting with

			sponsor and mentor.
Nathan Olson	Availability of test server	Start of second sprint	Email

4.3. Deliverables

<< Identify the major deliverables that this project is expected to produce. Assume the deliverables apply to all features or stories listed above unless indicated otherwise. Deliverables may include prototypes, documentation, software, etc. >>

Table 11: Deliverables

#	Deliverable
1	Study results <i>(if any)</i>
2	Code
3	Test and test results
4	Build process documents <i>(if any)</i>
5	Install process documents <i>(if any)</i>
6	Administrator or user manual <i>(if any)</i>
7	Postmortem document
8	Final report (final PowerPoint presentation, 3 minute video, and final sprint)

5. SYSTEM ANALYSIS AND DESIGN

This section describes in detail about the design part of the system.

5.1. Overall Description

<< Provide a more detailed, two to three paragraph description of the project. This description may include more technical details to describe the purpose of the project.

Example: This project is an attempt to apply data science and machine learning techniques to telemetry data from Microsoft ERP products in an attempt to anticipate user actions based on previous navigation and controls to create more efficient application navigation. Using Event Tracer for Windows, test teams are able to record millions of data points with information regarding actions that users have taken, forms they have navigated through, and the time data for each of those events. By combining the data and arranging it by session ID and time we can create a sort of roadmap of each user's actions in time order from the moment they start the application.

First this data must be cleaned and sorted using R statistical software. Then graph visualizations of the data (initially pertaining to the navigation path through the application) will be generated to allow us to view the dominant paths to specific modules and forms in the product, and give us an idea of where to start with machine learning. The statistics and visualization using R will be compiled into an R package to allow for easy documentation, and extensible use with various datasets. The visualization and statistics functions will provide basic insight into the data for any Program Manager, or someone not experienced in data science.

>>

5.2. Users and Roles

<< Provide a list and description of the different types of users or roles within the system. This may include different classes of users, such as administrator, instructor, student, etc. This list may also include autonomous agents that interact with the system as well. These may include users (or personas) that are used as part of any user stories produced for the project. >>

Example:

Table 12: Sample 12

User	Description
Developer	A capstone team member or mentor who is tasked with managing the test data, creating initial machine learning models, and ultimately generating a firm process for applying these techniques to future user data. This is used for sub-stories and task needed to fulfill the true end user use cases.
Microsoft Program Manager	A manager at Microsoft who is working on developing the ERP application who will be making design decisions based on the data analysis.
Dynamics AX User	An end user of the Microsoft ERP product who will be generating the data used and reaping the potential efficiency benefits from the data analysis when designing the application.

5.3. Design diagrams/Architecture/ UML diagrams/ Flow Charts/ E-R diagrams

<<Provide all the design diagrams that were created during the design phase of your project. Those who love coding may use a tool <https://liveuml.com/> to create

UML diagrams. You may refer <https://plantuml.com/> for commands/syntax of various diagrams. Some sample may be checked below:

- Use case: <https://liveuml.com/view/6024c067492d8164f7df07a3>
- Class: <https://liveuml.com/view/6024c3e8492d8164f7df07a4>

Some visual examples are in sections below:>>

5.3.1. Product Backlog Items

<<All features/services/functionalities are known as **Product Backlog Items (PBIs)**. When some features are picked from PBIs to develop in a SPRINT of two weeks, then it is known as **Sprint Backlog Items (SBIs)**. The features are then written as user stories, acceptance criteria, and tasks in a SPRINT.>>

Enlist all of your functionalities/requirements as user-stories in the format below:

As a <**user type**> I want <**actions**> So that <**achieved goal**>.

Examples:

- "As Max, I want to invite my friends, so we can enjoy this service together."
- "As Sascha, I want to organize my work, so I can feel more in control."
- "As a manager, I want to be able to understand my colleagues' progress, so I can better report our success and failures."
- "As a fitness enthusiast, I want to track my workouts and progress in an app so that I can monitor my improvements."
- "As a frequent shopper, I want to save my payment information so that I can check out quickly on future purchases."
- "As an online student, I want to receive notifications for upcoming classes so that I can easily manage my schedule."

5.3.2. Architecture Diagram

<< Provide the architecture diagram (may be a pattern selected) for your product>>.

5.3.3. Use Case Diagram

<< Provide any use-case diagrams that are being used as part of the project. Uniquely label each use case so that if necessary, it is easy to reference from other parts of the document. >>

Example: Restaurant system

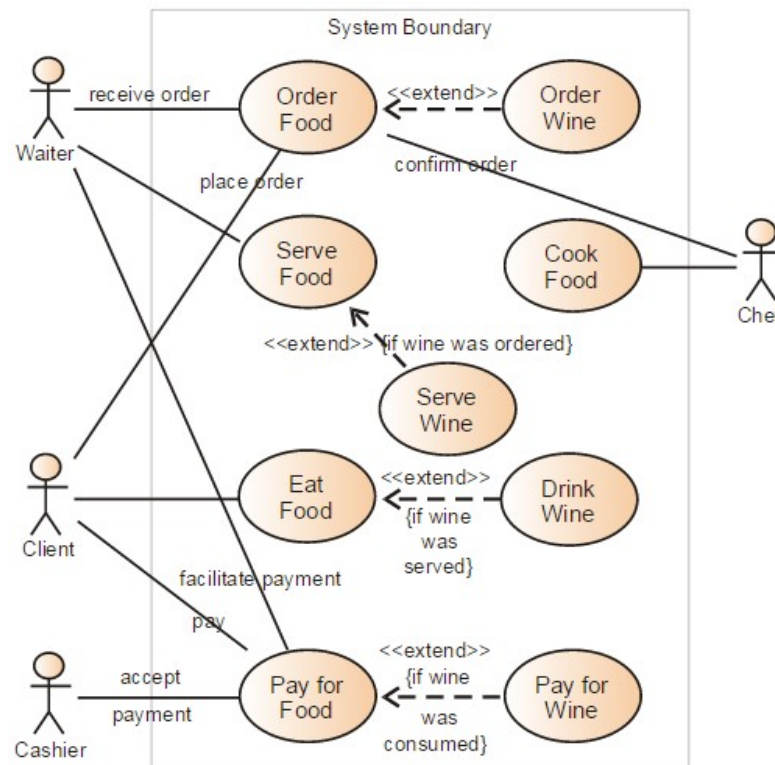


Figure 1: Sample use-case diagram

<<to insert caption for tables and figures, click on references on the top menu, under captions section → click insert caption → choose label as table

If you use a table, figure, or non-text item that is not your original design, you must cite the original source of the item. You may use an in-text citation in the text of the title or caption of the item, or you may include the citation as a footnote. Refer to the style manual of your discipline for more information about citations of non-text items.>>

5.3.4. Class Diagram

<< Include a class diagram for all classes to be designed. Optionally include major data elements of those classes and important methods and functions that will be used by other classes.>>

Example: Online Photo Collection

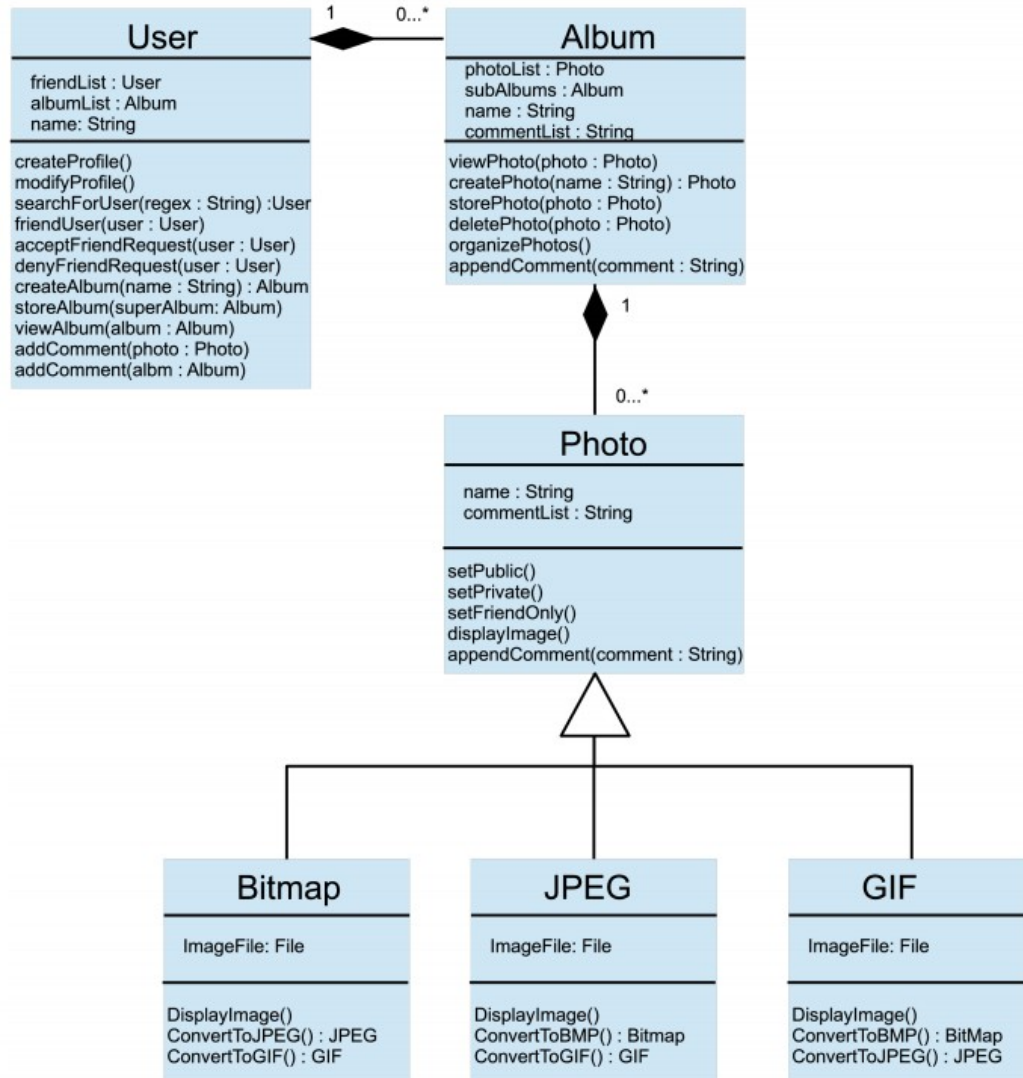


Figure 2: sample 2

5.3.5. Activity Diagrams

<< Include activity diagrams for important workflows in the program. At least one diagram should be included for the main workflow in the program. Optionally include labels that indicate which component is responsible for that part of the activity. Activity diagrams for components which perform complex tasks should also be included. >>

Example: View friend's photo album

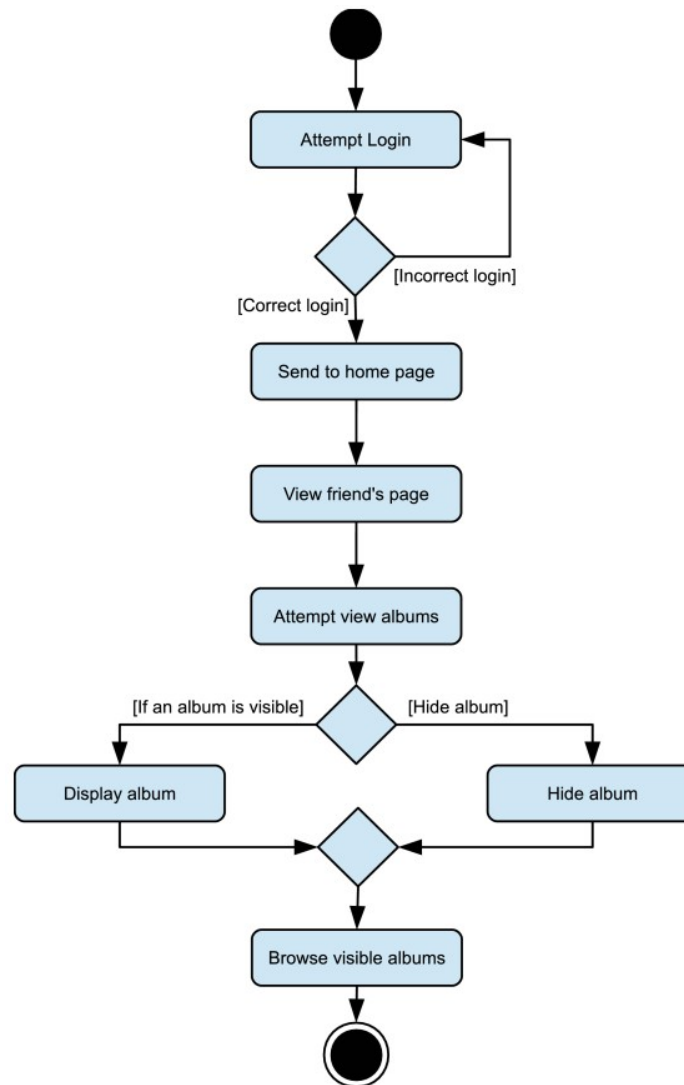


Figure 3: sample 3

5.3.6. Sequence Diagram

<< Include sequence diagrams for important functionality of the program to indicate control flow. These diagrams should include classes found in the class diagram and use the methods for those classes to show the interaction between them. >>

Example: Create new album

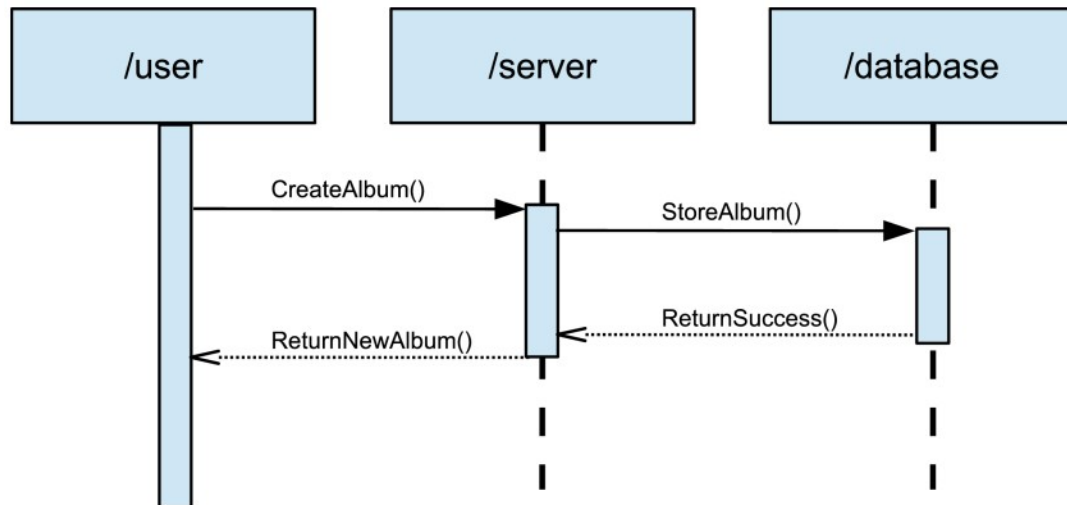


Figure 4: Sample 4

5.3.7. Data Architecture

<< Include any information or diagrams that provide details about databases, xml configuration files, or other data structures that are a part of the system. If a very specific format is required, it may be worthwhile to provide a more robust description or a detailed design such as a database schema. >>

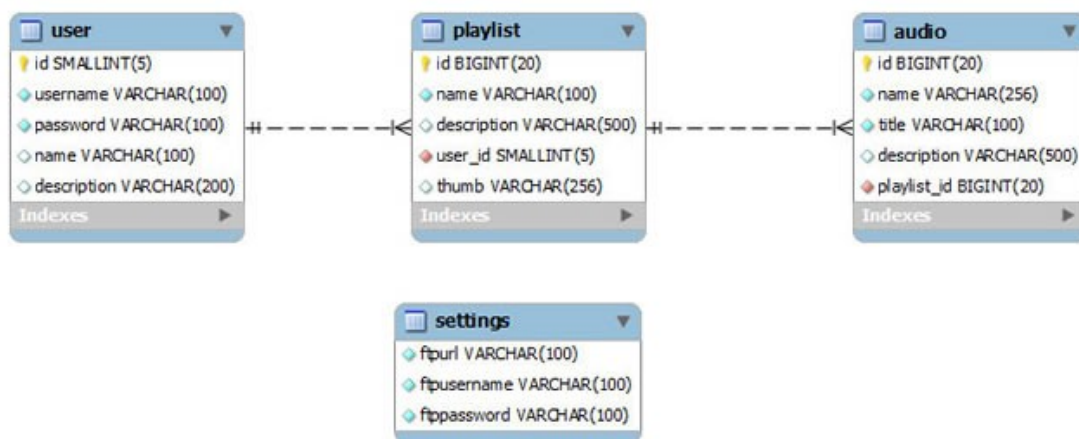


Figure 5: Sample 5

6. USER INTERFACE

6.1. UI Description

<< Provide a brief description of the UI that will be used in this program and how users will interact with the program.

Example: We are creating a research project using R and Azure ML. R is a statistical programming language that uses the R console, which will be the primary means of interacting with our code. This uses standard console UI, and it is not in the scope of our project to create a UI on top of that. The Azure ML machine learning model has its own built-in drag and drop UI for the different code modules we create, so there is no need to generate any sort of AI for that portion of the project either. >>

6.2. UI Mockup

<< Attach the screenshot of user interface. This can be a simple drawing that demonstrates key parts of the user interface.>>

Example:

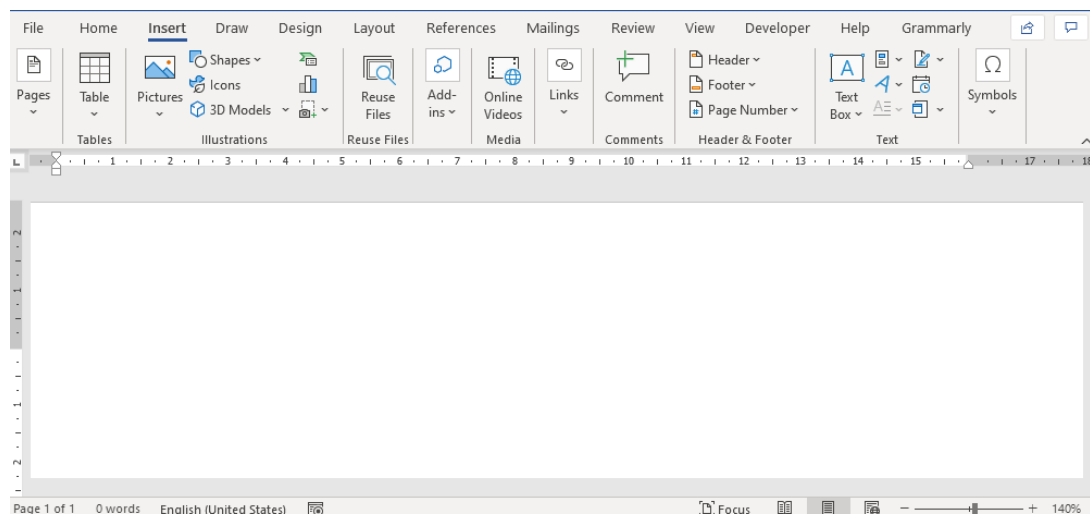


Figure 6: Sample 6

7. ALGORITHMS/PSEUDO CODE OF CORE FUNCTIONALITY

<<Include the proposed algorithm/pseudo code.>>

8. PROJECT CLOSURE

This section elucidates the overall lookup at the project and some of the future works that may enhance the solution.

8.1. Goals / Vision

<< Provide an update to the vision statement that was originally stated in the Project Initiation document.

Example: Our original goals for this project were to take telemetry data from Dynamics AX, analyze and visualize that data, then create machine learning models in Azure ML to predict user navigation based upon previous actions. Through the course of the project, these goals were altered so that the primary goal became creating a well-documented, extensible R language package that facilitated cleaning and importing telemetry data, and contained a variety of useful analysis and visualization functions to make the raw data more understandable. >>

8.2. Delivered Solution

<< Provide a high-level description of what was planned and what is being delivered.

Example: Our solution delivered primarily consisted of a fully documented, fully featured R language package that contained the functions for importing telemetry data, cleaning, separating, and isolating that data, then performing a variety of analysis, statistics, and visualizations on that data. This R package has standard R documentation, a full suite of unit tests, and an integrated manual and help documentation to allow anyone with a basic familiarity with the R environment to utilize our functions. >>

8.3. Remaining Work

<< Provide a short summary of what should be done next, ways of further improving the project, or any additional recommendations.

Example: We created a prototype of a web-based UI for our R package using Shiny in an attempt to make the functionality more accessible for those unfamiliar with R. There are a few minor functions implemented, and a basic UI setup, but a good deal of work would be required to flesh out this web app with all of the functions currently available in the R package, should Microsoft choose to pursue this UI overlay further. >>

REFERENCES

<<Would suggest you to download a tool (Mendeley: <https://www.mendeley.com>)

for automated citation and generation of references>>

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